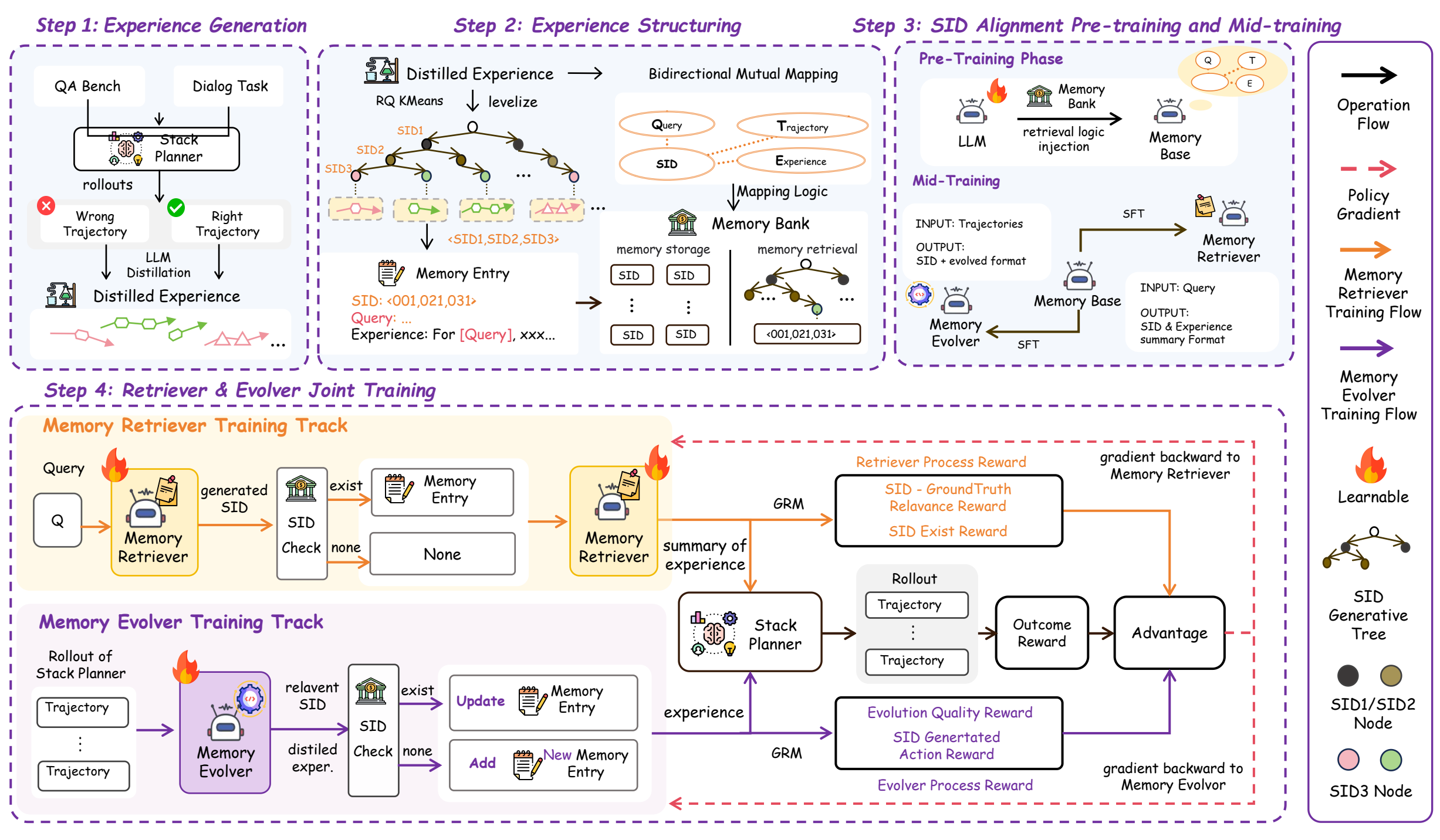

Step 1: Experience Generation
QA Bench
Dialog Task
Stack Planner
rollouts
Wrong Trajectory
Right Trajectory
LLM Distillation
Distilled Experience
Step 2: Experience Structuring
Distilled Experience
RQ KMeans
levelize
SID1
SID2
SID3
<SID1,SID2,SID3>
Memory Entry
SID: <001,021,031>
Query: ...
Experience: For [Query], xxx...
Bidirectional Mutual Mapping
Query
Trajectory
SID
Experience
Mapping Logic
Memory Bank
memory storage
memory retrieval
SID
<001,021,031>
Step 3: SID Alignment Pre-training and Mid-training
Pre-Training Phase
LLM
Memory Bank
retrieval logic injection
Memory Base
Q
T
E
Mid-Training
INPUT: Trajectories
OUTPUT:
SID + evolved format
SFT
Memory Retriever
Memory Base
INPUT: Query
OUTPUT:
SID & Experience summary Format
Memory Evolver
SFT
Step 4: Retriever & Evolver Joint Training
Memory Retriever Training Track
Query
Q
Memory Retriever
generated SID
SID Check
exist
none
Memory Entry
None
Memory Retriever
GRM
summary of experience
Retriever Process Reward
SID - GroundTruth Relavance Reward
SID Exist Reward
gradient backward to Memory Retriever
Memory Evolver Training Track
Rollout of Stack Planner
Trajectory
Trajectory
Memory Evolver
relavent SID
distiled exper.
SID Check
exist
none
Update
Memory Entry
Add
New Memory Entry
experience
GRM
Stack Planner
Rollout
Trajectory
Trajectory
Outcome Reward
Advantage
Evolution Quality Reward
SID Genertated Action Reward
Evolver Process Reward
gradient backward to Memory Evolvor
Operation Flow
Policy Gradient
Memory Retriever Training Flow
Memory Evolver Training Flow
Learnable
SID Generative Tree
SID1/SID2 Node
SID3 Node